

SURE-CHALLENGE: EVALUATING SPEECH EVIDENCE BEFORE SPEECH-LLM GENERATION

Mengzhe Geng

National Research Council Canada
Mengzhe.Geng@nrc-cnrc.gc.ca

ABSTRACT

Speech LLMs are usually graded after they answer, although an operating system first has to decide whether a waveform should be sent to the model. We define the Speech-Unsupported Rejection Evaluation Challenge (SURE-Challenge) for this admission step. The benchmark pairs LibriSpeech-derived transcription and first-word question answering with unsupported silence, colored noise, synthetic tones, and source-ambiguous babble under disjoint source splits. Front-end ablations use Qwen2-Audio; the selected energy-plus-Whisper-score rule is then replayed before six speech/audio LLMs. On the 474-row leakage-screened SURE-Extended test set, raw Qwen2-Audio rejects 15/204 unsupported inputs, whereas the fixed rule rejects 196/204 and leaves supported accuracy unchanged. External checks delimit this number: Common Voice retention drops as the Whisper-score threshold is tightened, and no-speed babble gives 18 to 24 rejected clips out of 54 across regenerated seeds. The result identifies a pre-generation error mode missed by answer-only scoring.

Index Terms— speech language models, audio hallucination, abstention, selective prediction, speech recognition confidence

1. INTRODUCTION

Speech-capable LLMs are now used for transcription, spoken instruction following, and audio question answering [1, 2]. In deployment, the first decision is made on the waveform itself: whether the input contains enough evidence to justify a model call. A clip may contain the requested utterance, pure background sound, music, a synthetic tone, or speech from a source that the prompt does not identify. Answer scoring begins after generation, but a fluent answer to an unsupported clip remains unsafe even when the text is well formed.

The operational decision is *speech-evidence filtering*. Given audio x and prompt q , a front end either admits the clip to a speech LLM or abstains without calling it. This check is only the input-side part of answer verification: usable speech-like evidence, semantic answerability, and speaker identity are distinct requirements. The setting connects selective prediction [3] with audio hallucination studies [4, 5, 6], but the error is measured before the generator can turn an unsupported waveform into a plausible response.

The benchmark instantiates the decision with source-disjoint splits at two sizes: SURE-Core for ablation and SURE-Extended for held-out evaluation. The public data and scripts are released with the paper¹. The experiments compare lightweight acoustic and ASR-confidence front ends, then replay one frozen decision rule before six speech/audio LLM backbones. External corpora probe

ordinary supported speech, real nonspeech and vocal audio, and overlapped speech outside the generator. These checks keep the main number in scope: a cheap fixed rule removes most synthetic unsupported calls, while threshold choice, multilingual decoding, and source ambiguity remain separate operating constraints.

2. RELATED WORK

Recent audio-language models attach acoustic encoders to language-model decoders and report strong results on speech and audio understanding [7, 1, 2, 8, 9, 10]. Most evaluations score the generated answer or mark unsupported content after generation [4, 5]. SURE-Challenge evaluates the preceding admission decision.

We draw front-end signals from ASR confidence estimation and voice activity detection, which reject uncertain hypotheses or non-speech segments [11, 12]. Whisper posteriors provide a related confidence signal [13, 6]. AudioSet taggers such as AST provide a speech-class baseline, although ESC-50, FSD50K, and VocalSound are close to their training domain [14, 15, 16, 17, 18]. Selective classification supplies the operating-point view for abstention [19, 3]. Speech LLMs add a case that these tools do not settle: an utterance may be intelligible and still unsupported when the prompt asks about a speaker or source that cannot be identified.

3. TASK AND DATA

Each SURE example contains audio, a prompt, an expected behavior label, and, for answerable cases, a reference transcript or answer. SURE-Core uses 40 LibriSpeech validation-clean utterances [20]; SURE-Extended uses 120 train-clean-100 utterances with the same generator. We split by source order and do not fit front-end parameters on train. Rows generated from the same source utterance are dependent, so the reported counts should not be read as independent per-row samples. Table 1 summarizes the resulting splits. The SURE-Extended test contains 270 supported and 210 unsupported examples; the main reported split has 474 rows after removing six held-out babble rows whose distractors wrap to train sources, with the matching dev rows removed before calibration.

The families separate recognition under acoustic perturbation from missing acoustic support. Supported rows include clean transcription, additive noise, filtering, reverberation, speed perturbation, and first-word question answering. Unsupported non-speech rows use length-matched silence, colored noise, and synthetic tones. Unsupported babble rows mix two or four off-source utterances and ask for the “main speaker.” The source utterance is absent, and the prompt gives no target identity, so selecting one speaker is incorrect. These rows test source attribution; the leakage-screened split

¹<https://github.com/MENGZHEGENG/sure-challenge>

Table 1. SURE benchmark composition. Counts are examples per split; the main SURE-Extended scores use 474 test rows after removing six cross-split babble cases.

Family	SURE-Core			SURE-Extended		
	Tr.	Dev	Test	Tr.	Dev	Test
Speech/QA	180	90	90	540	270	270
Noise/tone	100	50	50	300	150	150
Babble	40	20	20	120	60	60
Screened test	—	—	—	—	—	474

removes the dev and test cases whose distractors cross split boundaries.

The scoring rule follows the admission decision. Transcriptions are counted correct when normalized WER is at most 0.25, and first-word QA uses normalized exact containment. Unsupported rows require abstention or a clarification request; a generic tag such as [Music] does not satisfy a transcript prompt. Abstaining on supported speech is counted as an error. We report supported accuracy, unsupported rejection, WER on answered supported rows, and the number of downstream calls. External checks use Common Voice, FLEURS, ESC-50, FSD50K, VocalSound, MUSAN, and LibriCSS [21, 22, 14, 15, 16, 23, 24], plus AST speech tagging [17, 18] and 60 semantic-unanswerable speech prompts.

4. FRONT-END ADMISSION RULES

All admission rules use fixed signals extracted before the backbone runs. The front end receives audio x and prompt q , decides whether to admit the clip, and passes q downstream only after admission. The main rule removes extremely short or near-silent clips, decodes the rest with Whisper-small, and averages the maximum token probability at each step: $s(x) = T^{-1} \sum_t p_t$. It abstains when $s(x) < \tau$. We choose $\tau = 0.70$ on clean SURE-Extended dev by maximizing unsupported rejection under zero supported false rejects. Baselines use Silero VAD, AST AudioSet speech tags, Whisper no-speech probability, Whisper score alone, and simple fusions. For backbone-agnostic replay, the same frozen decisions are applied before Qwen2-Audio, Qwen2.5-Omni, Qwen-Audio-Chat, Audio Flamingo Next, Audio Flamingo 3, and MiniCPM-o [1, 2, 7, 8, 9, 10].

For source-ambiguous speech, we add audio-only overlap classifiers based on RMS, spectral, zero-crossing, and MFCC summaries. Thresholds are chosen on dev speakers with a supported-reject limit of 0.05 and then frozen for test.

5. RESULTS

5.1. Backbone Abstention Without a Front End

We first measure native refusal before adding a front end. Core ablations use held-out sources, and Extended uses the frozen thresholds from dev. On Core, raw unsupported rejection is 0.014 for Qwen2-Audio and 0.000 for Qwen-Audio-Chat and Qwen2.5-Omni; fixed-filter replay reaches 0.957. A self-abstention prompt rejects all 70 unsupported Core rows but leaves supported accuracy at only 0.033 for Qwen2-Audio and 0.156 for Qwen2.5-Omni, so the gain in refusal comes from suppressing valid generations as well as invalid ones. Table 2 reports the same frozen front-end replay across six backbones.

Five of the six backbones have WER below 0.05 on answered supported rows; Qwen-Audio-Chat is weaker at 0.202 WER, and all six seldom refuse unsupported inputs without a front end. For Qwen2-Audio, the fixed rule raises unsupported rejection from 15/204 to 196/204 and reduces downstream calls by 41%. All headline SURE-Extended rates use the screened 474-row test set. Raw Qwen2-Audio refusal remains low on both sizes: 1/70 on Core and 15/204 on the screened Extended split.

5.2. Front-End Controls

Table 3 compares matched front-end controls on the screened SURE-Extended test set.

The controls identify which cues drive the held-out result. Whisper no-speech probability behaves like raw Qwen2-Audio. AST and Silero reject all 150 synthetic non-speech rows and none of the babble rows. The Whisper score rejects 120/150 non-speech rows and 46/54 babble rows. Adding either the energy floor or Silero to the Whisper score produces the same 196/204 rejections on this split, because both additions cover the remaining synthetic non-speech rows and add no babble rejections beyond the score threshold. Fig. 1 and Table 4 test how the operating point behaves outside the SURE generator.

5.3. External Audio Checks

Table 5 reports two checks not shown in Fig. 1: multilingual decoder choice and semantically unanswerable speech prompts.

The external checks separate operating points that are tied on SURE-Extended. The main SURE rule retains 554/600 supported English clips and rejects 917/1160 unsupported external clips; this 1160-clip pool covers ESC-50, FSD50K music, MUSAN, VocalSound, and LibriCSS, with the regenerated no-speed babble row reported separately. VAD+score has nearly the same supported retention, 552/600, and higher unsupported rejection, 999/1160. AST and Silero also tie on SURE-Extended, but separate on real audio: AST rejects 997/1160 unsupported clips and retains 575/600 supported English clips, whereas Silero rejects 807/1160 and retains 596/600. A stricter Whisper threshold, $\tau = 0.90$, raises all-unsupported rejection to 0.964, including ESC-50 0.974, VocalSound 0.983, and LibriCSS 0.842, while reducing supported English retention to 0.505. The language and prompt checks show that language selection changes FLEURS retention by 153 clips, while the semantic-unanswerable prompts leave Qwen2-Audio’s refusal behavior unchanged after filtering. For LibriCSS, the prompt asks for the main speaker in an overlapped meeting segment, so rejection is scored as source-attribution abstention.

5.4. Source-Ambiguous Speech

Speech from the wrong source is the hardest unsupported condition. The fixed front end handles ordinary supported speech and synthetic non-speech cleanly, including 150/150 silence, noise, and tone cases. Babble behaves differently. The leakage-screened seed gives 46/54 overlap rejections; the original held-out seed had 60 overlap rows before six cross-split cases were removed. Across regenerated seeds, speed-perturbed babble gives 40–43/54 rejections, while no-speed babble gives 18–24/54. Raw Qwen2-Audio still almost never abstains, with 2/54 refusals on the no-speed, non-circular regeneration.

The overlap experiments mark the boundary of acoustic confidence. On the 60 pre-screened overlap rows, a logistic audio-only rule reaches dev/test ROC-AUC of 0.972/0.966 and catches 50/60

Table 2. Backbone-agnostic replay on leakage-screened SURE-Extended. Raw/SURE S. denote supported accuracy; Raw/SURE R. denote unsupported rejection.

Backbone	Raw S.	SURE S.	Raw R.	SURE R.	WER
Qwen2	.930	.930	.074	.961	.043
Omni	.967	.967	.000	.961	.035
Chat	.919	.919	.000	.961	.202
AF-Next	.956	.956	.000	.961	.045
AF3	.970	.970	.000	.961	.029
MiniCPM-o	.948	.948	.167	.961	.040

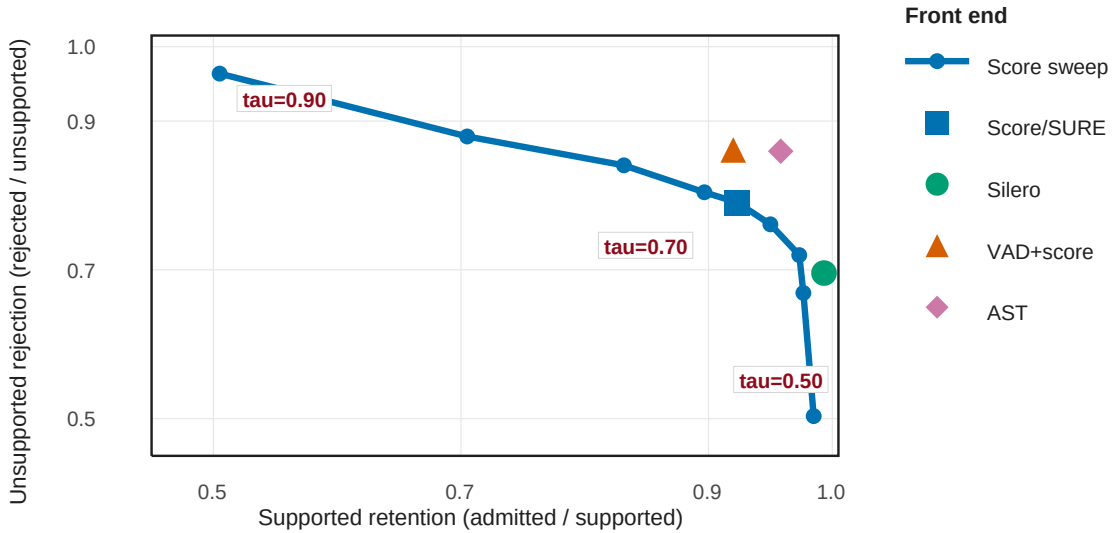


Fig. 1. Real-audio operating points. The line sweeps the Whisper-score threshold; markers show fixed front ends. The x-axis is supported retention (admitted supported clips divided by all supported clips), and the y-axis is unsupported rejection (rejected unsupported clips divided by all unsupported clips).

overlaps; it also rejects 10/270 supported examples. We keep this classifier out of the headline metric because it uses the pre-screened 60-row set and is trained against the synthetic babble construction. MFCC and ECAPA probes show the same LibriCSS trade-off. Confidence can flag unstable mixtures, but deciding that the requested speaker is absent requires source attribution.

6. LIMITATIONS AND IMPLICATIONS

The experiments separate recognizer errors from two admission errors: accepting audio with no speech evidence, and accepting speech whose source cannot be tied to the prompt. Larger backbones mainly affect recognition. They lower WER on admitted supported rows, yet they still answer many unsupported inputs unless an admission rule runs first. Prompting increases refusals only by discarding many valid answers. The front ends fail in different places: Silero preserves most supported speech and misses vocal and overlap cases; Whisper scoring is tunable but rejects Common Voice as the threshold rises; AST gives an AudioSet baseline with near-domain

bias. Because the rules run before generation, the same decisions can be replayed across backbones. On an H100 after model loading, support-gate, Whisper, and Qwen2-Audio paths average 0.012, 0.080, and 0.949 seconds per clip, respectively.

The main limitation is that audio-only admission cannot determine whether recognized speech answers a particular prompt. The semantic-unanswerable set and LibriCSS checks place that boundary in the results rather than leaving it as future work. Deployment also requires threshold selection matched to the cost of each error type. In our external checks, false accepts concentrate in overlap and vocal music, while false rejects appear mainly in Common Voice when the Whisper threshold is tightened. We did not evaluate dysarthric or child speech, and the multilingual result depends strongly on decoder language selection. External corpora are referenced by recipe without redistribution.

Table 3. Matched SURE-Extended controls. Gate R. denotes front-end rejection; Sys. R. also includes Qwen2 self-abstention.

Front end	Gate R.	Sys. R.	Supp.	Calls
Raw Qwen2	.000	.074	.930	474
No-sp. prob.	.000	.074	.930	474
Energy	.147	.221	.930	444
Silero	.735	.740	.930	324
AST	.735	.740	.919	321
Score	.814	.814	.930	308
V+S	.961	.961	.930	278
Energy+sc.	.961	.961	.930	278

Table 4. External checks. Supported rows report retention; unsupported rows report rejection. V+S denotes VAD+score, and – marks an AST slice not run.

Slice	<i>n</i>	SURE	Silero	V+S	AST
Common Voice	300	.847	.987	.840	.963
FLEURS-en	300	1.000	1.000	1.000	.953
ESC-50	500	.874	.958	.992	.990
FSD50K music	80	.812	.762	.938	.850
MUSAN	100	.950	.910	.980	.990
VocalSound	360	.858	.489	.886	.931
LibriCSS ovl.	120	.092	.000	.092	.000
No-speed bab.	54	.426	.000	.426	–

7. CONCLUSION

The evaluation asks whether a waveform should be sent to a speech LLM before answer scoring begins. Across six backbones, supported speech is usually transcribed correctly, but raw models rarely reject unsupported audio. On the held-out SURE split, the fixed acoustic rule raises Qwen2-Audio unsupported rejection from 0.074 to 0.961 without reducing supported accuracy. The external rows identify where that scalar result does not transfer unchanged: equal SURE scores can have different external behavior, and overlap requires source attribution. For Qwen2-Audio, answer-only scoring would leave 181 unsupported decisions uncounated.

8. REFERENCES

- [1] Yunfei Chu et al., “Qwen2-audio technical report,” *arXiv preprint arXiv:2407.10759*, 2024.
- [2] Qwen Team, “Qwen2.5-omni technical report,” *arXiv preprint arXiv:2503.20215*, 2025.
- [3] Yonatan Geifman and Ran El-Yaniv, “Selective classification for deep neural networks,” in *Advances in Neural Information Processing Systems*, 2017, pp. 4878–4887.
- [4] Xize Cheng et al., “AHa-Bench: Benchmarking audio hallucinations in large audio-language models,” in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2025.
- [5] Feiyu Zhao et al., “HalluAudio: A comprehensive benchmark for hallucination detection in large audio-language models,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026.

Table 5. Language and prompt checks. FLEURS reports multilingual retention under English-fixed versus automatic Whisper decoding; semantic rows report Qwen2-Audio refusal.

Check	<i>n</i>	Base	Adjusted
FLEURS multi.	200	.235	1.000
Semantic unans.	60	.767	.767

- [6] Mateusz Barański, Jan Jasiński, Julitta Bartolewska, Stanisław Kacprzak, Marcin Witkowski, and Konrad Kowalczyk, “Investigation of whisper ASR hallucinations induced by non-speech audio,” *arXiv preprint arXiv:2501.11378*, 2025.
- [7] Yunfei Chu et al., “Qwen-audio: Advancing universal audio understanding via unified large-scale audio-language models,” *arXiv preprint arXiv:2311.07919*, 2023.
- [8] Sreyan Ghosh et al., “Audio flamingo next: Next-generation open audio-language models for speech, sound, and music,” *arXiv preprint arXiv:2604.10905*, 2026.
- [9] NVIDIA, “Audio Flamingo 3: Advancing audio intelligence with fully open large audio language models,” Model card, 2026.
- [10] OpenBMB Team, “MiniCPM-o 2.6: A GPT-4o-level MLLM for vision, speech and multimodal live streaming on your phone,” Technical report and model card, 2025.
- [11] Hui Jiang, “Confidence measures for speech recognition: A survey,” *Speech Communication*, vol. 45, no. 4, pp. 455–470, 2005.
- [12] Silero Team, “Silero VAD: Pre-trained enterprise-grade voice activity detector,” <https://github.com/snakers4/silero-vad>, 2024.
- [13] Alec Radford et al., “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning*, 2023, pp. 28492–28518.
- [14] Karol J. Piczak, “ESC: Dataset for environmental sound classification,” in *Proceedings of the 23rd ACM International Conference on Multimedia*, 2015, pp. 1015–1018.
- [15] Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra, “FSD50K: An open dataset of human-labeled sound events,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2022.
- [16] Yuan Gong, Jin Yu, and James Glass, “VocalSound: A dataset for improving human vocal sounds recognition,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2022.
- [17] Jort F. Gemmeke, Daniel P. W. Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R. Channing Moore, Manoj Plakal, and Marvin Ritter, “Audio Set: An ontology and human-labeled dataset for audio events,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2017, pp. 776–780.
- [18] Yuan Gong, Yu-An Chung, and James Glass, “AST: Audio spectrogram transformer,” in *Proceedings of Interspeech*, 2021, pp. 571–575.
- [19] C. K. Chow, “On optimum recognition error and reject trade-off,” *IEEE Transactions on Information Theory*, vol. 16, no. 1, pp. 41–46, 1970.

- [20] Vassil Panayotov et al., “LibriSpeech: An ASR corpus based on public domain audio books,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2015, pp. 5206–5210.
- [21] Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis M. Tyers, and Gregor Weber, “Common voice: A massively-multilingual speech corpus,” in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 2020, pp. 4218–4222.
- [22] Alexis Conneau et al., “FLEURS: Few-shot learning evaluation of universal representations of speech,” *arXiv preprint arXiv:2205.12446*, 2022.
- [23] David Snyder, Guoguo Chen, and Daniel Povey, “MUSAN: A music, speech, and noise corpus,” *arXiv preprint arXiv:1510.08484*, 2015.
- [24] Zhuo Chen et al., “Continuous speech separation: Dataset and analysis,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, 2020, pp. 7284–7288.